

MAKURAZAKI, Japan

WMO: 478310

Lat: 31.2719N

Lon: 130.2931E

Elev: 103

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.6	36.7	22.5	17.0	41.7	24.9	19.0	42.2	25.8	50.0	23.9	48.6	9.1	0	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	89.7	78.6	88.3	78.3	87.1	78.0	81.0	86.1	80.1	85.2	79.4	84.5	10.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.6	154.4	84.2	78.8	150.0	83.6	78.0	145.9	83.1	44.8	86.3	43.9	85.3	43.1	84.8	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.2	21.1	18.9	30.8	92.6	1.6	1.5	29.6	93.7	28.6	94.5	27.7	95.4	26.6	96.4		
			27.9	81.7	1.5	1.4	26.8	82.7	25.9	83.5	25.1	84.3	24.0	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	47.7	49.7	54.3	61.4	68.2	73.9	81.0	82.5	78.0	69.9	60.5	51.6
	DBStd	12.84	5.49	6.15	5.65	4.95	3.51	3.61	2.74	2.23	3.71	4.89	6.00	6.01
	HDD50	260	110	75	22	1	0	0	0	0	0	0	1	51
	HDD65	2023	536	430	331	127	11	0	0	0	0	13	159	416
	CDD50	5730	39	65	156	343	565	717	962	1006	841	618	317	101
	CDD65	2019	0	0	1	19	111	267	497	541	391	166	25	1
	CDH74	17185	0	0	0	14	266	1358	5166	6166	3394	782	39	0
	CDH80	4947	0	0	0	0	9	120	1568	2282	892	76	0	0
Wind	WSAvg	9.3	10.3	10.4	10.5	9.8	8.7	8.0	8.4	8.9	8.7	8.7	9.1	10.0
Precipitation	PrecAvg	92.6	3.7	4.5	6.5	7.6	8.0	21.1	11.8	8.0	8.7	4.0	4.7	4.0
	PrecMax	149.7	7.6	10.6	11.9	18.4	14.1	46.3	39.2	21.7	23.2	14.8	10.4	7.0
	PrecMin	61.3	1.1	1.4	1.9	2.0	1.9	7.8	0.9	1.7	1.1	0.6	1.9	0.4
	PrecStd	20.7	1.7	2.4	2.4	4.1	3.4	7.9	8.0	5.0	5.7	3.3	2.2	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	68.7	71.0	76.2	81.3	84.9	91.3	92.1	89.0	84.5	77.4	70.7
		MCWB	61.7	63.4	62.5	65.7	67.7	77.6	78.4	78.9	77.2	75.0	70.3	63.6
	2%	DB	63.7	66.4	68.4	73.2	78.6	83.0	89.2	90.3	87.3	82.1	75.0	67.4
		MCWB	58.4	61.5	61.5	64.5	67.3	76.2	78.4	78.8	76.9	73.4	66.8	61.1
	5%	DB	60.8	64.0	66.7	71.5	76.8	81.3	87.8	88.9	86.1	80.2	73.0	64.9
		MCWB	55.0	58.8	60.0	63.8	67.2	75.1	78.5	78.6	76.2	71.4	65.5	58.4
	10%	DB	57.6	60.9	64.7	70.0	75.2	79.9	86.4	87.7	84.6	78.4	70.8	62.2
		MCWB	51.5	55.1	58.2	62.8	66.9	74.5	78.3	78.4	75.6	69.6	63.7	55.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	65.1	66.2	70.9	74.7	79.7	81.9	82.6	81.0	78.0	72.7	65.9
		MCDB	66.2	67.7	68.0	73.7	77.7	83.0	87.3	87.9	85.5	82.2	75.9	68.4
	2%	WB	59.4	62.2	63.7	68.6	72.5	78.3	80.9	81.3	79.7	75.6	69.3	62.4
		MCDB	63.2	65.7	67.2	70.9	75.2	81.1	85.9	86.6	84.4	80.3	73.2	66.3
	5%	WB	56.1	59.3	61.5	66.4	70.9	77.3	80.0	80.4	78.5	73.7	67.1	59.6
		MCDB	59.9	63.5	65.5	69.5	73.9	80.1	84.9	85.8	83.4	78.8	71.4	63.8
	10%	WB	52.1	55.8	59.2	64.5	69.3	76.2	79.2	79.6	77.3	71.4	64.9	56.6
		MCDB	57.1	60.1	63.9	68.6	73.0	79.1	84.1	85.2	82.7	76.5	69.6	61.3
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.6	14.0	13.7	13.0	9.6	9.5	10.5	11.8	13.6	14.1	13.6
		MCDBR	16.7	15.6	15.4	14.7	15.3	10.7	11.7	12.3	12.6	14.0	14.6	15.8
		MCWBR	13.3	13.1	11.3	9.2	8.1	5.4	4.8	4.9	5.4	7.3	9.2	11.6
	5% WB	MCDBR	14.5	13.5	12.8	10.2	9.2	7.3	9.7	10.3	10.1	10.8	11.3	13.5
		MCWBR	13.3	12.9	12.1	9.2	7.6	5.0	4.7	4.9	5.3	6.8	9.1	11.6
Clear-Sky Solar Irradiance	taub	0.440	0.475	0.529	0.544	0.547	0.546	0.520	0.512	0.500	0.473	0.439	0.431	
	taud	2.081	2.017	1.887	1.905	1.964	2.060	2.165	2.194	2.177	2.161	2.176	2.106	
	Ebn at Noon	240	245	241	243	243	242	248	249	246	243	239	235	
	Edh at Noon	43	50	61	62	59	53	48	46	45	43	39	40	
All-Sky Solar Radiation	RadAvg	707	920	1256	1494	1561	1303	1759	1761	1445	1186	853	679	
	RadStd	70	94	89	146	174	199	191	117	149	133	85	70	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page